Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Anaphylaxis

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Under the Auspices of the IAP Action Plan 2022

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Anaphylaxis

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Definition

An acute clinical syndrome characterized by severe, life threatening, generalized type 1 hypersensitivity reaction, usually caused by exposure to a foreign substance leading to mast cell degranulation and release of chemical mediators.

Diagnostic Criteria

Anaphylaxis is highly likely when **any one** of the following **3 criteria** are fulfilled:

1. **Acute onset**[a] of an illness with **involvement of the skin, mucosal tissue, or both**[b] AND AT LEAST ONE OF THE FOLLOWING:
   a. Respiratory compromise[c]
   b. Circulatory compromise[d]

2. **Two or more** of the following that occur **acutely**[a] **after exposure to a likely allergen** for that patient: a. Involvement of the skin-mucosal tissue[b]
   b. Respiratory compromise[c]
   c. Circulatory compromise[d]
   d. Persistent gastrointestinal (GI) symptoms[e]

3. **Acutely after exposure to known allergen** for that patient, **reduced BP**-defined as **low SBP for age**[f] or **> 30% decrease in SBP**

[a]**Acute onset:** Onset within minutes to hours
[b]**Involvement of the skin, mucosal tissue:** Includes generalized hives or urticaria, pruritus, flushing, swollen lips-tongue-uvula
[c]**Respiratory compromise:** Includes dyspnea, wheeze-bronchospasm, low PEFR, stridor, hypoxemia
[d]**Circulatory compromise:** Includes low BP or associated symptoms of end-organ dysfunction like hypotonia, collapse, syncope, incontinence
[e]**GI symptoms:** Crampy abdominal pain, vomiting
[f]**Low systolic blood pressure (SBP)** for children is defined as: < 60 mm Hg from birth to 1 month, < 70 mm Hg from 1 month to 1 year, < (70 mm Hg + [2 × age]) from 1 to 10 years, < 90 mm Hg for 11 years and above

Modified from National Institute of Allergy and Infectious Diseases/Food Allergy and Anaphylaxis Network (NIAID/FAAN) criteria for diagnosis of anaphylaxis in the emergency department (ED).

Precipitating Factors

- ☑ **Enteral ingestion:** Drugs (Antibiotics, NSAIDs, Aspirin), Food
- ☑ **Parenteral administration**: Drugs (Antibiotics, Paralytic agents, Opioids, Propofol), Insect bites and stings, Contrast media, Vaccines, Blood products
- ☑ **Physical contact:** Latex
- ☑ **Others:** Exercise in cases of FDEIA (Food-dependent exercise-induced anaphylaxis)

Lab Parameters

Anaphylaxis is a clinical diagnosis and no lab parameters are required for diagnosis. However, serial measurements of serum tryptase done at the acute stage and later can help to rule out systemic mastocytosis or pseudo anaphylaxis.

Differential Diagnosis

Features	*Anaphylaxis*	*Vaso Vagal Attack*
Onset	In Minutes	In Seconds
Common Triggers	Medications, Insect Bite, Exercise, Milk, Shellfish	Pregnancy, Dehydration, Extreme Pain/Fear/Stress/Heart Block
Skin	Red, Hot and Wheals with Itchy Rashes	Pale and Cold
Respiratory System	Adventitious Sounds Like Wheeze/Stridor	Normal
Cardiovascular System	Tachycardia with Hypotension	Bradycardia with Hypotension
GI System	Nausea/Vomiting	Non-Specific
Prone Position	No Improvement	Good Response
Treatment	IM Adrenaline Inj	Treat the Cause

Management Algorithm

Clinical Suspect

Patient satisfying any of the 3 criterias (as mentioned above) in presence of a known trigger or precipitating factor

Initial Management (First 1 min)

- ☑ Keep Supine Position (**Sitting Position If respiratory distress/nausea/vomiting)**
- ☑ Assess Airway, Breathing, Circulation, heart rate, BP & SpO_2
- ☑ Provide O_2 @ 10-15 L/min by NRBM (If respiratory distress/shock)
- ☑ Identify and Remove the Allergic Trigger if possible (e.g., insect sting)
- ☑ Administer **1st Dose of IM** Epinephrine @ 0.01mg/kg/dose (1mg/ml of 1:1000 dilution) at anterolateral aspect of thigh
- ☑ Can be repeated at 5 to 10 mins interval as needed

1 to 5 min

Respiratory Distress

- ☑ Sitting Position
- ☑ Provide High Flow O_2 by HFNC and plan for intubation
- ☑ If Stridor, 1st nebulized epinephrine @ 0.5 mL/kg
- ☑ If Wheeze, 1st nebulized salbutamol @ 0.15 mg/kg

Hypotension/Poor Perfusion/Loss of Consciousness

- ☑ Supine Position
- ☑ IV/IO Assess to be established
- ☑ 1st Bolus of NS @ 20 mL/kg by IV/IO Rapid Push Technique

5 to 10 min

If No Improvement give 2nd Dose of IM Epinephrine

- ☑ IV/IO Assess to be Established (If not done earlier
- ☑ Prepare for Difficult Intubation

For Respiratory Distress

- ☑ 2nd nebulized epinephrine @ 0.5 mL/kg for Stridor; 2nd nebulized salbutamol @ 0.15 mg/kg for Wheeze

Hypotension/Poor Perfusion/Loss of Consciousness

- ☑ 2nd Bolus of NS @ 20ml/kg by IV/IO Rapid Push Technique
- ☑ Prepare for IV Epinephrine

Alert Pediatric ICU/ Tertiary Care Centre

10 to 20 min

If no Improvement give 3rd Dose of IM Epinephrine

For Respiratory Distress

- ☑ 3rd nebulized epinephrine @ 0.5 mL/kg for Stridor; 3rd nebulized salbutamol @ 0.15 mg/kg for Wheeze
- ☑ Proceed for Intubation

Hypotension/Poor Perfusion/Loss of Consciousness

- ☑ Start giving **IV** Epinephrine @ 0.05 mcg/kg/min with titrate by 0.02 mcg/kg/min up to effect

Transfer to Pediatric ICU/ Tertiary Care Centre

If still no Improvement Suspect Refractory Anaphylaxis

- ☑ Start IV Norepinephrine Infusion (persistent hypotension) @ 0.05 mcg/kg/min with titrate by 0.02mcg/kg/min up to effect (Max: 2 mcg/kg/min)
- ☑ IV Glucagon Bolus (persistent anaphylaxis/patients on beta blockers) @ 20 – 30 mcg/kg/dose (Max: 1 mg) over 5 mins followed by 5 to15 mcg/min titrated till achievement of clinical effects

Manage as per Pediatric ICU protocols

Monitoring & Follow Up

- ☑ Delayed Anaphylaxis may be due to biphasic reaction (1 to 20% cases) which generally occurs within 1 to 72 hours (mostly within 10 hours) after initial improvement or protracted variety which is the severe persistent anaphylactic symptoms for 24 to 36 hours despite aggressive treatment.
- ☑ Monitoring of vitals (BP, SpO2) along with Respiratory Symptoms (stridor or wheeze) should be done over minutes during first 30 mins and then hourly for next 24 hour and then 2 hourly for next 48 hour.
- ☑ Follow up of patients should be done for at least 4 hours after last IM Epinephrine injection and in severe hospitalized cases for minimum 3 days.

Indication of Hospitalization

- ☑ Tachycardia, Hypotension (SBP in mm Hg: < 70 + Age in Years × 2)
- ☑ Stridor, Wheeze, Tachypnoea (1 to 5 Year - ≥ 40 / min; > 5 Year - ≥ 30 / min; >10 Year - ≥ 20 /min)
- ☑ SpO2 < 95% in Room Air
- ☑ History of Severe Protracted Anaphylaxis
- ☑ Comorbidities (asthma, arrhythmia, systemic mastocytosis)
- ☑ Live in remote area or present late in the evening

Home Action Plan

- ☑ Hives with Itching is the earliest (most common) sign though it may not happen in few cases
- ☑ EpiPen *(Green colour - 0.15mg / inject; Yellow colour - 0.3mg / inject)* is ideal but due its limited resources in India, home prepared IM Epinephrine Inj (0.01mg / kg of 1:1000 dilution) in 1ml Syringe over anterolateral aspect of thigh may be a practical option.
- ☑ It can be kept for 3 months at room temperature with protection from light. (expiry date should be written for future reference)
- ☑ Call Your State Emergency Number (National Emergency No – 112)

Key Learning Points

- ☑ Anaphylaxis is a sudden, severe, life-threatening allergic reaction to an allergen with multi system involvement.
- ☑ Early identification and management are the corner stone in good outcome.
- ☑ IM Epinephrine is the Drug of Choice.
- ☑ Follow up of patients for minimum of 3 days due to biphasic nature of anaphylaxis.
- ☑ Education of Parents about its early recognition and appropriate management is the need of hour.

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Difficulties in Breastfeeding

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Difficulties in Breastfeeding

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Inverted/Flat Nipples

Nipple protractility is important than shape and size of nipple. If nipple stands prominent when pinched at base, it is a flat nipple. If nipple goes in, then it is a retracted nipple. Flat nipple does not pose any problems, as baby suckles at breast not nipple.

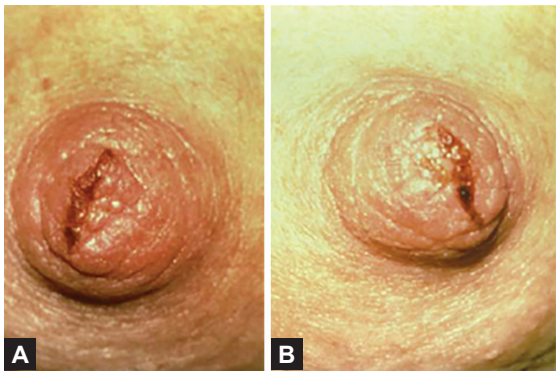
Postnatal Management

Antenatal management of flat/retracted nipple is not beneficial:

- ☑ Management starts at delivery.
- ☑ Build mother's confidence that with patience and perseverance, she will succeed.
- ☑ Help mother to position baby early, trying different positions. Provide skin-to-skin contact.
- ☑ Help her to make nipple stand out more (method shown in **Fig. 2B**).
- ☑ For retracted nipple, use syringing initially (Kesari method). Assure that problem will resolve gradually.

Sore Nipples

Causes: Incorrect positioning and attachment of baby, nipple sucking, frequent use of soap and water on breast, and fungal infection of nipple.



Treatment

- ☑ Continue breastfeeding with correct position and attachment.
- ☑ Expose the nipple to air between feeds and apply hind milk to the nipple after breastfeeding.
- ☑ Counsel mother to avoid frequent washing of breasts.
- ☑ If fungal infection occurs, apply medicine on the nipple and inside the baby's mouth.

A Good attachment **B** Poor attachment

C Good positioning

Modified cradle position: Sitting, the mother cradles the baby in her arm. Her hand supports the baby's head and neck.

Breast Engorgement

It occurs due to milk accumulation. Breasts are swollen, hard, areola gets overstretched, and milk does not flow well.

- ☑ *Causes*: Delayed initiation of breastfeeding, poor attachment to breasts, plenty of milk, infrequent removal of milk, and restriction of length of feeds.
- ☑ *Prevention*: Providing support to breastfeeding soon after delivery to all mothers, ensure good attachment, and encourage unrestricted breastfeeding.

Treatment

Apply warm water packs for not >15 minutes. Paracetamol to relieve pain.

Breast Abscess

The mother may have high-grade fever, pain in breast, and overlying skin is warm and red.

Treatment

It includes incision and drainage of abscess, analgesics, and antibiotics. Breastfeeding must be continued from the other breast.

Not Enough Milk

Milk is adequate if baby passes urine at least six times a day and gains weight adequately.

Causes

Not breastfeeding frequently, poor position, poor attachment, too short or hurried breastfeeds, and breast engorgement or mastitis.

Treatment

More frequent feeds, especially at night, proper attachment to breast, care of painful condition, back massage for stimulating lactation, and build mother's confidence.

Vaccination

All vaccines given to lactating woman are safe for women and their infants, except for smallpox and yellow fever vaccines.

Medications

Only a small proportion of medications are contraindicated in breastfeeding mothers. The reader should refer to LactMed (available freely at https://www.ncbi.nlm.nih.gov/books/NBK501922/?report=classic) to obtain the most current data on an individual medication.

Working Mother

She should exclusively breastfeed for 6 months. She may be encouraged to carry the baby to a work place/crèche wherever such facility exists. She can express her milk in clean container which can be stored at room temperature for up to 4–6 hours, in the refrigerator for 4 days and in the freezer for about 6 months. Expressed milk can be given with spoon.

Cesarean Delivery

Start breastfeeding as early as possible and preferably within 1 hour of birth. Help mother to find a comfortable position to breastfeed. She may feed in her supine position (on first day), side-lying position (on second day), or sitting position (day three onwards).

Supine position on day 1.

Side-lying position on day 2.

Infant having Cleft Lip and/or Cleft Palate

- ☑ Each child with this problem needs individual evaluation.
- ☑ Mothers and family should be counseled about likely breastfeeding success.
- ☑ If direct breastfeeding is unlikely then expressed breast milk can be given by cup/spoon.
- ☑ Semi-upright position is recommended to reduce nasal regurgitation and reflux of breast milk into the eustachian tubes.

Football position on right breast for unilateral cleft lip on left side and vice versa.

Dancer's hand position for hypotonic infant.

Hypotonic Infant

- ☑ Breastfeeding should be initiated as soon as the infant is stable. Encourage skin-to-skin contact, closely monitor latch, position, and suck of the baby.
- ☑ Consider alternative modes of feeding such as a cup, spoon, or ghokarnam or orogastric feeding if the infant is unable to breastfeed or sustain adequate suckling.
- ☑ Regular follow-up for weight gain.

Contraindications to Breastfeeding

- ☑ Mother using an illicit drug.
- ☑ Refrain from breastfeed or giving expressed milk if mother is infected with human T-cell lymphotropic virus type I or type II or Ebola virus infection, infants with classic galactosemia, maple syrup urine disease, and phenylketonuria.
- ☑ Temporarily avoid breastfeeding as well as her expressed milk if she is infected with untreated brucellosis and when she is on medications like anticancer or radioactive compounds.
- ☑ If mother has herpes simplex virus lesions on the breast or having active untreated sputum positive tuberculosis then she should avoid direct breastfeeding, but expressed milk can be given.

Breastfeeding in Preterm and Very Low Birth Weight Neonates

- ☑ Mode of feeding should be based on maturity.
- ☑ First choice of milk is mother's own milk (MoM), followed by pasteurized donor milk or preterm formula.
- ☑ *Enteral feeding*: Start minimal enteral feeding, preferably within 24 hours of life with 10–20 mL/kg/day. Fortification once reaches 100 mL/kg/day at 1 sachet of 1 g in 25 mL breast milk to improve calorie and protein intake.
- ☑ *Multivitamins*: 1 mL/kg/day, calcium (110–220 mg/kg/day), phosphorus (55–110 mg/kg/day), and iron (4 mg/kg/day).
- ☑ Monitor the growth of the newborn regularly.

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